has not really lost the information carried by the stuff that has fallen into the black hole; it is simply locked within a region of space that we rational beings choose to avoid at all costs? Prior to Hawking's realization that black holes are not completely black, the answer to these questions was yes. But once Hawking informed the world that black holes radiate, the story changed. Radiation carries energy and so, as a black hole radiates, its mass slowly decreases—it slowly evaporates. As it does so, the distance from the center of the hole to the event horizon slowly shrinks, and as this shroud recedes, regions of space that were previously cut off re-enter the cosmic arena. Now our philosophical musings must face the music: Does the information contained in the things swallowed by the black hole—the data we imagined existing within the black hole's interior—re-emerge as the black hole evaporates? This is the information required for quantum determinism to hold, and so this question goes to the heart of whether black holes imbue the evolution of our universe with an even deeper element of happenstance.

As of this writing, there is no consensus among physicists regarding the answer to this question. For many years, Hawking has strongly claimed that the information does not re-emerge—that black holes destroy information thereby "introducing a new level of uncertainty into physics, over and above the usual uncertainty associated with quantum theory"9 In fact, Hawking, together with Kip Thorne of the California Institute of Technology, has a bet with John Preskill, also of the California Institute of Technology, regarding what happens to the information captured by a black hole: Hawking and Thorne bet that the information is forever lost, while Preskill has taken the opposite position and bet that the information re-emerges as the black hole radiates and shrinks. The wager? Information itself: "The loser(s) will reward the winner(s) with an encyclopedia of the winner's choice."

The bet remains unsettled, but Hawking has recently acknowledged that the newfound understanding of black holes from string theory, as discussed above, shows that there might be a way for the information to re-emerge.10 The new idea is that for the kind of black holes studied by Strominger and Vafa, and by many other physicists since their initial paper, information can be stored and